

Intel Prescott 3.2 GHz processor

A little too new

Intel has finally announced the launch of its new Prescott line of processors. The Prescotts are the first desktop processors to be based on the 90 nm manufacturing process.

The new processor core competes directly with AMD's Athlon 64 line, but will also help phase out the Northwood-based Pentium 4 processors by the end of 2004. The new processors will be available from 2.8 GHz to 3.4 GHz, for now—Intel estimates that they will cross the 4 GHz mark by the end of 2004. For processor speeds above 3.4 GHz, a new chipset and socket is under way. Current Prescott processors have the same pin count as Northwood, and are compatible with the Socket 423 motherboard. Most motherboards supporting 800 FSB processors will be able to take the 3.4 GHz Prescott, though a BIOS upgrade from the board manufacturer might be required in certain cases.

The new manufacturing process will help Intel better the yield per silicon wafer, thus lowering the price per processor. However, it will be some time before Intel can pass on the benefits to consumers.

The new Prescott processors have a host of new developments at the manufacturing level, such as the use of 'strained silicon'—atomic level changes in the silicon lattice—and smaller gate transistor lengths. All of this helps make the transistor work faster, and allows for more transistors to be cramped on to a chip.

Other improvements, such as the inclusion of seven layers of copper interconnect, as

opposed to 6 on a normal Northwood Pentium 4, help in designing a complex logical block, but without increasing the core area. One particular disadvantage of the 90 nm technique is current leakage, which results in higher voltage requirement and thus, higher heat dissipation.

Unlike the Pentium4 EE, where Intel just added an additional L3 cache to the existing Northwood core, the Prescott is an entire new processor, with heavy revisions done to the NetBurst micro-architecture—the term introduced for Pentium 4 design architecture. The idea behind this architecture was to increase CPU performance by increasing the clock frequency; but increasing the frequency calls for lengthening of the execution pipeline—a line-up of 'to do things'. This again means improving the branch predicting unit, so that it results in proper predictions, saving

applications.

Our test system comprised an Intel 875PBZ motherboard, ATI Radeon 9600 Pro Graphic Card, 512 MB DDR400 RAM and a 120 GB Seagate SATA hard drive. Windows XP with SP1 and all necessary drivers were loaded.

In the Content Creation 2004 test, the Prescott could only outperform the Athlon XP 3200+. Even the older Northwood-based Pentium 4 3.2 GHz beat it—this is due to the longer pipeline in the Prescott, which increases the latency. This is only temporary, and will vanish as processor frequencies increase. The Business Winstone test saw similar results.

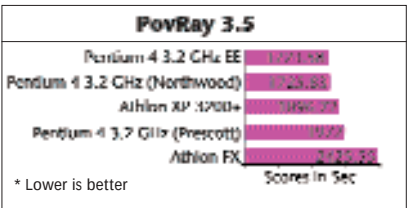
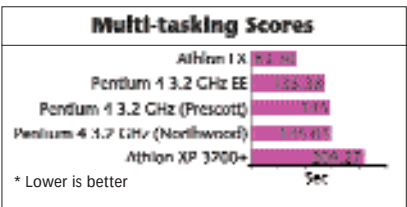
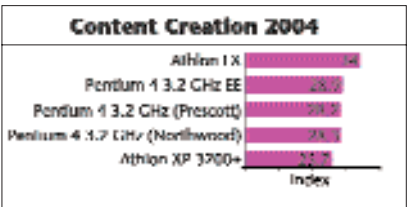
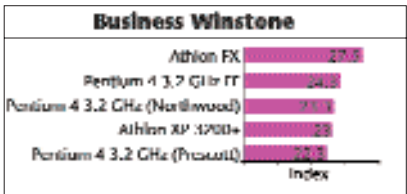
In UT2003, the Prescott lost

to the 3.2EE by 3 fps—scoring 80.39 at a resolution of 640 x 480. In the other gaming tests that we put it through, it was always a hair's breadth behind the 3.2EE.

In the PovRay 3.5 OpenGL test—a rendering benchmark—the Prescott lost badly to the 3.2EE and the Athlon XP 3200+. Though the Athlon FX was the slowest, we suspect this is due to PovRay's problem with the 64-bit Athlon FX processor.

In the Super Pi test, the Prescott returned better scores than the 3.2EE and Athlon XP 3200+. This can be attributed to the improved execution unit within the processor.

The Prescott 3.2 GHz is



time lost due to wrong predictions. Thus, an improved branch predicting unit is under the hoods of the Prescott.

The most noticeable change is the increase in cache memories. The L1 cache has gone from 8 KB to 16 KB, and the L2 has also doubled to 1 MB. The introduction of 13 new SIMD instructions also improves graphic rendering

temptingly priced at Rs 14,000, but for now, is too immature a line to recommend. It's a good performer, but doesn't really offer any significant advantages, yet. We suggest you wait for the Prescott's migration to the different socket and chipset, where the advantages will be more visible.

Price: Rs 14,000
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SYSTEM REQUIREMENTS

Socket 478, 1 MB L2 Cache, 90 nm fabrication Process, 800 MHz FSB, HT enabled

Intel Prescott 3.2 GHz processor	A-
Performance	4.0
Features	4.0
Build quality	4.0
Value for money	4.0